

Appendix A
(Normative Appendix)
Data Coding

A.1 Data Format Description

XXXXXX.XX represents the integer and decimal places of the measured value or stored value; NNNNNN.NN represents the integer and decimal places of the set value; YY represents year; MM represents month; DD represents day; WW represents weekday; hh represents hour; mm represents minute; ss represents second; unless otherwise specified, all numbers are expressed in double digits. decimal number representation.

A.2 Data Identification Code Table

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13	D12	D11	D10				Read	Write	
00	00	00 01 ... 04 FF	00	XXXXXX.XX	4	kWh	*		(Current) combined active total energy (Current) Combined Active Rate 1 Electric Energy ... (Current) Combined Active Rate 4 Electric Energy (Current) Combined active energy data block
00	01	00 01 ... 04 FF	00	XXXXXX.XX	4	kWh	*		(Current) total forward active energy (Current) Forward active rate 1 Electric energy ... (Current) Forward active rate 4 Electric energy (Current) forward active energy data block
00	02	00 01 ... 04 FF	00	XXXXXX.XX	4	kWh	*		(Current) total reverse active energy (Current) Reverse active rate 1 Electric energy ... (Current) Reverse active rate 4 Electric energy (Current) reverse active energy data block
00	03	00 01 ... 04 FF	00	XXXXXX.XX	4	1.5 oz.	*		(Current) Combined reactive power 1 total power (Current) Combined Reactive Power 1 Rate 1 Energy ... (Current) Combined Reactive Power 1 Rate 4 Electric Energy (Current) Combined Reactive 1 Energy Data Block
00	04	00 01 ... 04 FF	00	XXXXXX.XX	4	1.5 oz.	*		(Current) Combined reactive power 2 total power (Current) Combined Reactive Power 2 Rate 1 Electric Energy ... (Current) Combined Reactive Power 2 Rate 4 Electric Energy (Current) Combined Reactive 2 Energy Data Block
00	05	00 01 ... 04 FF	00	XXXXXX.XX	4	1.5 oz.	*		(Current) total reactive energy in the first quadrant (Current) First Quadrant Reactive Rate 1 Electric Energy ... (Current) First Quadrant Reactive Rate 4 Electric Energy (Current) First quadrant reactive energy data block
00	06	00 01 ... 04 FF	00	XXXXXX.XX	4	1.5 oz.	*		(Current) Total reactive energy in the second quadrant (Current) Second Quadrant Reactive Power Rate 1 Electricity ... (Current) Second Quadrant Reactive Rate 4 Electric Energy (Current) Second quadrant reactive energy data block
00	07	00 01 ... 04	00	XXXXXX.XX	4	1.5 oz.	*		(Current) Total reactive energy in the third quadrant (Current) Third Quadrant Reactive Power Rate 1 Electricity ... (Current) Third Quadrant Reactive Rate 4 Electric Energy

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12		D11 D10					Read	Write	
		FF							(Current) Third quadrant reactive energy data block
00	08	00 01 ... 04 FF	00	XXXXXX.XX	4	1.5 oz.	*		(Current) Total reactive energy in the fourth quadrant (Current) Fourth Quadrant Reactive Power Rate 1 Electricity ... (Current) Fourth Quadrant Reactive Rate 4 Electric Energy (Current) Fourth quadrant reactive energy data block
00	15 16 17 18 19 1A 1B 1C	00	00	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		(Current) Phase A forward active energy (Current) Phase A reverse active energy (Current) Phase A combined reactive power 1 (Current) Phase A combined reactive 2 energy (Current) Phase A first quadrant reactive energy (Current) Phase A second quadrant reactive energy (Current) Phase A third quadrant reactive energy (Current) Phase A fourth quadrant reactive energy
00	29 2A 2B 2C 2D 2E 2F 30 31 32	00	00	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. kWh kWh	*		(Current) B phase forward active energy (Current) Phase B reverse active energy (Current) B phase combined reactive power 1 (Current) B phase combined reactive 2 energy (Current) Phase B first quadrant reactive energy (Current) Phase B second quadrant reactive energy (Current) Phase B third quadrant reactive energy (Current) Phase B fourth quadrant reactive energy
00	3D 3E 3F 40 41 42 43 44 45 46	00	00	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. kWh kWh	*		(Current) Phase C forward active energy (Current) Phase C reverse active energy (Current) Phase C combined reactive power 1 (Current) Phase C combined reactive 2 energy (Current) Phase C first quadrant reactive energy (Current) Phase C second quadrant reactive energy (Current) Phase C third quadrant reactive energy (Current) Phase C fourth quadrant reactive energy
00	00	00 01 ... 04 FF	01	XXXXXX.XX	4	kWh	*		(Previous settlement day) Combined active energy (Previous settlement day) Combined active rate 1 Electric energy ... (Previous settlement date) Combined active rate 4 Electric energy (Previous settlement date) Combined active energy data block
00	01	00 01 ... 04 FF	01	XXXXXX.XX	4	kWh	*		(Previous settlement date) Total forward active energy (Previous settlement day) Forward active power rate 1 Electric energy ... (Previous settlement date) Forward active power rate 4 Electric energy (Previous settlement day) Forward active energy data block
00	02	00 01 ... 04 FF	01	XXXXXX.XX	4	kWh	*		(Previous settlement day) Reverse active energy (Previous settlement day) Reverse active power rate 1 Electric energy ... (Previous settlement day) Reverse active power rate 4 Electricity (Previous settlement day) Reverse active energy data block
00	03	00 01 ...	01	XXXXXX.XX	4	1.5 oz.	*		(Previous settlement day) Combined reactive power 1 total energy (Previous settlement date) Combined reactive power 1 rate 1 energy ...

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13	D12	D11	D10				Read	Write	
		04 FF							(Previous settlement date) Combined reactive power 1 rate 4 energy (Previous settlement day) Combined reactive power 1 energy data block
00	04	00 01 ... 04 FF	01	XXXXXX.XX	4	1.5 oz.	*		(Previous settlement day) Combined reactive power 2 total energy (Previous settlement date) Combined reactive power 2 rate 1 electric energy ... (Previous settlement date) Combined reactive power 2 rate 4 energy (Previous settlement day) Combined reactive power 2 data block
00	05	00 01 ... 04 FF	01	XXXXXX.XX	4	1.5 oz.	*		(Previous settlement date) Total reactive energy in the first quadrant (Previous settlement day) Reactive power rate in the first quadrant 1 Electric energy ... (Previous settlement date) Reactive power rate in the first quadrant 4 Electricity (Previous settlement day) Reactive energy data block in the first quadrant
00	06	00 01 ... 04 FF	01	XXXXXX.XX	4	1.5 oz.	*		(Previous settlement date) Total reactive energy in the second quadrant (Previous settlement date) Second quadrant reactive power rate 1 Electricity ... (Previous settlement date) Second quadrant reactive power rate 4 Electricity (Previous settlement day) Second quadrant reactive energy data block
00	07	00 01 ... 04 FF	01	XXXXXX.XX	4	1.5 oz.	*		Total reactive energy in the third quadrant (previous settlement date) (Previous settlement date) Third quadrant reactive power rate 1 Electricity ... (Previous settlement date) Third quadrant reactive power rate 4 Electricity (Previous settlement day) Third quadrant reactive energy data block
00	08	00 01 ... 04 FF	01	XXXXXX.XX	4	1.5 oz.	*		(Previous settlement date) Total reactive energy in the fourth quadrant (Previous settlement date) Fourth quadrant reactive power rate 1 Electricity ... (Previous settlement date) Fourth quadrant reactive power rate 4 Electricity (Previous settlement day) Fourth quadrant reactive energy data block
00	15 16 17 18 19 1A 1B 1C	00	01	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		(Previous settlement date) Phase A forward active energy (Previous settlement date) Phase A reverse active energy (Previous settlement date) Phase A combined reactive power 1 (Previous settlement date) Phase A combined reactive power 2 (Previous settlement date) Phase A first quadrant reactive energy (Previous settlement date) Phase A second quadrant reactive energy (Previous settlement date) Phase A third quadrant reactive energy (Previous settlement date) Phase A fourth quadrant reactive energy
00	29 2A 2B 2C 2D 2E 2F 30	00	01	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		(Previous settlement date) Phase B forward active energy (Previous settlement date) Phase B reverse active energy (Previous settlement date) Phase B combined reactive power 1 (Previous settlement date) Phase B combined reactive power 2 (Previous settlement date) Reactive energy in the first quadrant of phase B (Previous settlement date) Phase B second quadrant reactive energy (Previous settlement date) Phase B third quadrant reactive energy (Previous settlement date) Phase B fourth quadrant reactive energy
00	3D 3E 3F 40 41 42 43 44 45 46	00	01	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. kWh kWh	*		(Previous settlement date) Phase C forward active energy (Previous settlement day) Phase C reverse active energy (Previous settlement date) Phase C combined reactive power 1 (Previous settlement date) Phase C combined reactive power 2 (Previous settlement date) Reactive energy in the first quadrant of phase C (Previous settlement date) Phase C second quadrant reactive energy (Previous settlement date) Phase C third quadrant reactive energy (Previous settlement date) Reactive energy in the fourth quadrant of phase C

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12		D11 D10					Read	Write	
00				...					
00	00	00 01 ... 04 FF	0C	XXXXXX.XX	4	kWh	*		(Previous 12th settlement day) Combined active energy (12th settlement day) Combined active power rate 1 Electricity ... (12th settlement day) Combined active rate 4 electricity (Previous 12th settlement day) Combined active energy data block
00	01	00 01 ... 04 FF	0C	XXXXXX.XX	4	kWh	*		(Previous 12th settlement day) Total forward active energy (Previous 12th settlement day) Forward active power rate 1 Electric energy ... (12th settlement day) Forward active rate 4 electric energy (Previous 12th settlement day) Forward active energy data block
00	02	00 01 ... 04 FF	0C	XXXXXX.XX	4	kWh	*		(Previous 12 settlement days) Reverse active energy (12th settlement day) Reverse active power rate 1 Electricity ... (12 settlement days before) Reverse active rate 4 electricity (Previous 12th settlement day) Reverse active energy data block
00	03	00 01 ... 04 FF	0C	XXXXXX.XX	4	1.5 oz.	*		(Previous 12th settlement day) Combined reactive power 1 total power (12 settlement days before) Combined reactive power 1 rate 1 electric energy ... (12 settlement days before) Combined reactive power 1 rate 4 energy (Previous 12th settlement day) Combined reactive power 1 energy data block
00	04	00 01 ... 04 FF	0C	XXXXXX.XX	4	1.5 oz.	*		(Previous 12 settlement days) Combined reactive power 2 total power (12 settlement days before) Combined reactive power 2 rate 1 electric energy ... (12 settlement days before) Combined reactive power 2 rate 4 energy (Previous 12 settlement days) Combined reactive 2 power data block
00	05	00 01 ... 04 FF	0C	XXXXXX.XX	4	1.5 oz.	*		(Previous 12th settlement day) Total reactive energy in the first quadrant (12 settlement days before) First quadrant reactive power rate 1 Electricity ... (12 settlement days before) First quadrant reactive power rate 4 electric energy (Previous 12th settlement day) Reactive energy data block in the first quadrant
00	06	00 01 ... 04 FF	0C	XXXXXX.XX	4	1.5 oz.	*		(Previous 12th settlement day) Total reactive energy in the second quadrant (12 settlement days before) Second quadrant reactive power rate 1 Electricity ... (12th settlement day) Second quadrant reactive power rate 4 electricity (Previous 12th settlement day) Second quadrant reactive energy data block
00	07	00 01 ... 04 FF	0C	XXXXXX.XX	4	1.5 oz.	*		Total reactive energy in the third quadrant (last 12 settlement days) (12th settlement day) Third quadrant reactive power rate 1 Electricity ... (12th settlement day) Third quadrant reactive power rate 4 electricity (Previous 12th settlement day) Third quadrant reactive energy data block
00	08	00 01 ... 04 FF	0C	XXXXXX.XX	4	1.5 oz.	*		(Previous 12th settlement day) Total reactive energy in the fourth quadrant (12 settlement days before) Fourth Quadrant Reactive Power Rate 1 Electricity ... (12th settlement day) Fourth quadrant reactive power rate 4 electric energy (Previous 12th settlement day) Fourth quadrant reactive energy data block
00	15 16 17 18 19 1A	00	0C	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		(Previous 12th settlement day) Phase A forward active energy (Previous 12th settlement day) Phase A reverse active energy (Previous 12th settlement day) Phase A combined reactive power 1 (Previous 12th settlement day) Phase A combined reactive 2 energy (Previous 12th settlement day) Phase A first quadrant reactive energy (Previous 12th settlement day) Phase A second quadrant reactive energy

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13	D12	D11	D10				Read	Write	
	1B 1C					1.5 oz. 1.5 oz.			(Previous 12th settlement date) Phase A third quadrant reactive energy (Previous 12th settlement day) Phase A fourth quadrant reactive energy
00	29 2A 2B 2C 2D 2E 2F 30	00	0C	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		(Previous 12th settlement day) Phase B forward active energy (Previous 12th settlement day) Phase B reverse active energy (Previous 12th settlement day) Phase B combined reactive power 1 (Previous 12th settlement day) Phase B combined reactive 2 energy (Previous 12th settlement day) Reactive energy in the first quadrant of phase B (Previous 12th settlement day) Phase B second quadrant reactive energy (Previous 12th settlement day) Phase B third quadrant reactive energy (Previous 12th settlement day) Phase B fourth quadrant reactive energy
00	3D 3E 3F 40 41 42 43 44	00	0C	XXXXXX.XX	4	kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		(Previous 12th settlement day) Phase C forward active energy (Previous 12th settlement day) Phase C reverse active energy (Previous 12th settlement day) Phase C combined reactive power 1 (Previous 12th settlement day) Phase C combined reactive 2 energy (Previous 12th settlement day) Reactive energy in the first quadrant of phase C (Previous 12th settlement day) Phase C second quadrant reactive energy (Previous 12th settlement day) Phase C third quadrant reactive energy (Previous 12th settlement day) Phase C fourth quadrant reactive energy
Note 1: The most significant bit of the combined active and reactive energy is the sign bit, 0 for positive and 1 for negative. Value range: 0.00 to 799999.99. Note 2: ZZ represents any value listed in this byte. ZZ cannot be FF. Note 3: The definition of the four quadrants for electric energy measurement is shown in Appendix D. Note 4: When transmitting a certain daily energy data block for settlement, the rate energy contained in the data block shall be based on the actual set rate number.									

Table A.2 Maximum demand and occurrence time data identification code table

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13	D12	D 11	D10				Read	Write	
01	01	00 01 ... 04 FF	00	XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute	*		(Current) Maximum total forward active demand and occurrence time (Current) Forward Active Power Rate 1 Maximum Demand and Time of occurrence ... (Current) Forward Active Rate 4 Maximum Demand and Time of occurrence (Current) Maximum forward active power demand and when it occurs Intermediate data block
01	02	00 01 ... 04	00	XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute	*		(Current) Maximum total demand and occurrence of reverse active power time (Current) Reverse Active Power Rate 1 Maximum Demand and Time of occurrence ... (Current) Reverse Active Power Rate 4 Maximum Demand and Time of occurrence
01	03	00 01 ... 04	00	XX.XXXX YYMMDDhhmm	8	left Year Month Day Time Minute	*		(Current) Combined reactive power 1 Total maximum demand and generation Birth time (Current) Combined reactive power 1 rate 1 maximum demand and Time of occurrence ... (Current) Combined reactive power 1 rate 4 maximum demand and

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13	D12	D D11 D10	D10				Read	Write	
		FF							Time of occurrence (Current) Combined reactive power 1 maximum demand and occurrence Time data block
01	04	00 01 ... 04 FF	00	XX.XXXX YYMMDDhhmm	8	left Year Month Day Time Minute	*		(Current) Combined reactive power 2 Total maximum demand and generation Birth time (Current) Combined Reactive Power 2 Rate 1 Maximum Demand and Time of occurrence ... (Current) Combined Reactive Power 2 Rate 4 Maximum Demand and Time of occurrence (Current) Combined reactive power 2 maximum demand and occurrence Time data block
01	15 16 17 18	00 00		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute	*		(Current) Maximum forward active power demand and generation of phase A Birth time (Current) Maximum demand and generation of active power in phase A Birth time (Current) Phase A combined reactive power 1 maximum demand and generation Birth time (Current) Phase A combined reactive power 2 maximum demand and generation Birth time
01	29 2A 2B 2C	00 00		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute	*		(Current) Maximum forward active power demand and generation of phase B Birth time (Current) Maximum demand and generation of active power in phase B Birth time (Current) Phase B combined reactive power 1 maximum demand and generation Birth time (Current) Phase B combined reactive power 2 maximum demand and generation Birth time
01	3D 3E 3F 40	00 00		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute	*		(Current) Maximum forward active power demand and generation of phase C Birth time (Current) Phase C maximum reverse active power demand and generation Birth time (Current) Phase C combined reactive power 1 maximum demand and generation Birth time (Current) Phase C combined reactive power 2 maximum demand and generation Birth time
01	01	00 01 ... 04 FF	01	XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute	*		(Previous settlement day) Maximum total forward active power demand and time of occurrence (Previous settlement day) Forward active power rate 1 Maximum required Quantity and time of occurrence ... (Previous settlement day) Forward active power rate 4 Maximum required Quantity and time of occurrence (Previous settlement day) Maximum forward active power demand and Occurrence time data block
01	02	00 01 ...	01	XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute	*		(Previous settlement day) Maximum total demand for reverse active power and time of occurrence (Previous settlement day) Reverse active power rate 1 Maximum required Quantity and time of occurrence ...

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
DI3	DI2	D I1	DI0				Read	Write	
		04 FF							(Previous settlement day) Reverse active power rate 4 Maximum required Quantity and time of occurrence (Previous settlement day) Maximum reverse active power demand and Occurrence time data block
01	03	00 01 ... 04 FF	01	XX.XXXX YYMMDDhhmm	8	left Year Month Day Time Minute	*		(Previous settlement day) Combined reactive power 1 total maximum demand and time of occurrence (Last settlement day) Combined reactive power 1 rate 1 maximum Demand and occurrence time ... (Last settlement day) Combined reactive power 1 rate 4 maximum Demand and occurrence time (Previous settlement day) Combined reactive power 1 maximum demand and Occurrence time data block
01	04	00 01 ... 04 FF	01	XX.XXXX YYMMDDhhmm	8	left Year Month Day Time Minute	*		(Previous settlement day) Combined reactive power 2 total maximum demand and time of occurrence (Previous settlement day) Combined reactive power 2 rate 1 maximum Demand and occurrence time ... (Previous settlement day) Combined reactive power 2 rate 4 maximum Demand and occurrence time (Previous settlement day) Combined reactive power 2 maximum demand and Occurrence time data block
01	15 16 17 18	00 01		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute	*		(Previous settlement date) Maximum forward active power demand of phase A and time of occurrence (Previous settlement day) Maximum demand for active power in the reverse direction of phase A and time of occurrence (Previous settlement date) Maximum reactive power requirement of phase A Quantity and time of occurrence (Previous settlement date) Maximum reactive power requirement of phase A combination 2 Quantity and time of occurrence
01	29 2A 2B 2C 2D	00 01		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute	*		(Previous settlement date) Maximum forward active power demand of phase B and time of occurrence (Previous settlement day) Maximum demand for active power in the reverse direction of phase B and time of occurrence (Previous settlement day) Maximum reactive power requirement of phase B combination 1 Quantity and time of occurrence (Previous settlement date) Maximum reactive power requirement of phase B combination 2 Quantity and time of occurrence
01	3D 3E 3F 40	00 01		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute	*		(Previous settlement date) Maximum forward active power demand of phase C and time of occurrence (Previous settlement day) Maximum demand for reverse active power in phase C and time of occurrence (Previous settlement date) Maximum reactive power requirement of phase C combination 1 Quantity and time of occurrence (Previous settlement date) Maximum reactive power requirement of phase C combination 2 Quantity and time of occurrence
01				...	...	...	...		...
01	01	00 0C		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute	*		(12 settlement days before) Maximum total forward active power demand and time of occurrence

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
DI3	DI2	D DI1	DI0				Read	Write	
		01 ... 04 FF							(12th settlement day) Forward active power rate 1 Maximum demand and occurrence time ... (Previous 12 settlement days) Forward active power rate 4 Maximum demand and occurrence time (Previous 12 settlement days) Forward active power maximum demand and occurrence time data block
01	02	00 01 ... 04 FF	0C	XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute	*		(Previous 12 settlement days) Reverse active power total maximum demand and occurrence time (Previous 12 settlement days) Reverse active power rate 1 Maximum demand and occurrence time ... (12 settlement days ago) Reverse active power rate 4 Maximum demand and occurrence time (12 settlement days ago) Reverse active power maximum demand and occurrence time data block
01	03	00 01 ... 04 FF	0C	XX.XXXX YYMMDDhhmm	8	left Year Month Day Time Minute	*		(12 settlement days ago) Combined reactive power 1 Total maximum demand and occurrence time (12 settlement days ago) Combined reactive power 1 Rate 1 Maximum High demand and time of occurrence ... (Previous 12 settlement days) Combined reactive power 1, rate 4, maximum demand and occurrence time (Previous 12 settlement days) Combined reactive power 1, maximum demand and occurrence
01	04	00 01 ... 04 FF	0C	XX.XXXX YYMMDDhhmm	8	kvar year month day hour minute	*		time data block (Previous 12 settlement days) Combined reactive power 2, total maximum demand and occurrence time (Previous 12 settlement days) Combined reactive power 2, rate 1, maximum demand and occurrence time ... (12th settlement day before) Combined reactive power 2 rate 4 maximum demand and occurrence time (12th settlement day before) Combined reactive power 2 maximum demand and occurrence time data block
01	15 16 17 18	00 0C		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Hour Minute kW Year Month Day Time Minute left Year Month Day Hour Minute kvar Year Month Day Time Minute	*		(Previous 12th settlement date) Maximum forward active power demand of phase A and time of occurrence (Previous 12th settlement date) Maximum reverse active power demand of phase A and time of occurrence (Previous 12th settlement date) Maximum combined reactive power 1 demand of phase A and time of occurrence (Previous 12th settlement date) Maximum combined reactive power 2 demand of phase A and time of occurrence
01	29 2A 2B 2C	00 0C		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Hour Minute kW Year Month Day Hour Minute left Year Month Day Hour Minute kvar Year Month Day Time Minute	*		(Previous 12th settlement date) Maximum forward active demand of phase B and time of occurrence (Previous 12th settlement date) Maximum reverse active demand of phase B and time of occurrence (Previous 12th settlement date) Maximum combined reactive power 1 demand of phase B and time of occurrence (Previous 12th settlement date) Maximum combined reactive power 2 demand of phase B and time of occurrence

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
DI3	DI2	D I1	DI0				Read	Write	
01	3D 3E 3F 40	00 0C		XX.XXXX YYMMDDhhmm	8	kW Year Month Day Time Minute kW Year Month Day Time Minute left Year Month Day Time Minute left Year Month Day Time Minute left	*		(12th settlement day before) Maximum forward active power required for phase C Quantity and time of occurrence (12th settlement day) Maximum reverse active power demand for phase C Quantity and time of occurrence (12 settlement days before) Phase C combined reactive power 1 maximum Demand and occurrence time (12 settlement days before) Phase C combined reactive power 2 maximum Demand and occurrence time
Note 1: The most significant bit of the combined maximum reactive demand is the sign bit, which is 0 for positive and 1 for negative. The value range is 0.0000 to 79.0000. Note 2: When transmitting the data block of the maximum demand and occurrence time of a settlement day, the maximum demand and occurrence time of the rate contained in the data block shall be based on the actual set rate number. allow.									

Table A.3 Variable data identification coding table

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
DI3	DI2	DI1	DI0				Read	Write	
02	01	01 02 03 FF	00	XXX.X	2	In	*		Phase A voltage Phase B voltage Phase C voltage Voltage data block
02	02	01 02 03 FF	00	XXX.XXX	3	A	*		Phase A current Phase B current Phase C current Current data block
02	03	00 01 02 03 FF	00	XX.XXXX	3	kW	*		Instantaneous total active power Instantaneous phase A active power Instantaneous phase B active power Instantaneous phase C active power Instantaneous active power data block
02	04	00 01 02 03 FF	00	XX.XXXX	3	left	*		Instantaneous total reactive power Instantaneous phase A reactive power Instantaneous B phase reactive power Instantaneous C phase reactive power Instantaneous reactive power data block
02	06	00 01 02 03 FF	00	X.XXX	2		*		Total power factor Phase A power factor Phase B power factor Phase C power factor Power factor data block
02	07	01 02 03 FF	00	XXX.X	2	°	*		Phase A angle Phase B angle C phase angle Phase angle data block
02	80	00	01 02 03 04 05 07 08 09 0A	XXX.XXX XX.XX XX.XXXX XX.XXXX XX.XXXX XXX.X XX.XX XX.XX XXXXXXXXX	3 2 3 3 3 2 2 2 4	A Hz kW kW left y In In point	*		Neutral current Grid frequency Total average active power in one minute Current active power demand Current reactive power demand Temperature inside the table Clock battery voltage (internal) Power outage meter reading battery voltage (external) Internal battery operating time

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
DI3	DI2	DI1	DI0				Read	Write	
Note 1: For a three-phase three-wire meter, the voltage for phase A is Uab, the voltage for phase B is 0, and the voltage for phase C is Ucb. The current for phase A is Ia, the voltage for phase B is 0, and the voltage for phase C is Ic. The power factor for phase A is the cosine of the angle between Uab and Ia, the voltage for phase B is 0, and the voltage for phase C is the cosine of the angle between Ucb and Ic. The phase angle for phase A is the angle between Uab and Ia, the voltage for phase B is 0, and the voltage for phase C is the angle between Ucb and Ic. Note 2: The highest digit of instantaneous power and current demand indicates the direction, 0 is positive, 1 is negative, and phase B of three-phase three-wire is 0. Value range: 0.0000~79.9999. Note 3: The highest digit of temperature in the table, 0, indicates above zero, and 1 indicates below zero. Value range: 0.0~799.9. Note 4: The phase angle measurement range is 0~360 degrees. Note 5: The current active demand, current reactive demand, and current apparent demand are the average power in the most recent period. Note 6: The highest digit of current indicates the direction, 0 is positive, 1 is negative, and the value range: 0.000~799.999. The highest digit of power factor indicates the direction, 0 is positive, 1 is negative, and the value range: 0.000~1.000.									

Table A.4 Event record data identification code table

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12	D11 D10						Read	Write	
03	11	00	00	XXXXXX	3 times*				Total power failures
03	11	00	01	YYMMDDhhmmss YYMMDDhhmmss	12		*		(Last) Power failure occurrence and end time
			...						...
			0A						(Last 10 times) The total number of times the power failure
03 30	00		00	XXXXXX	3		*		occurs and ends
03 30	00		01	YYMMDDhhmmss	6		*		(Last 1) Programming record: The
				C0C1C2C3	4				first 10 data
				XXXXXXXX	4×10				identification codes
									programmed by the operator code at the time of occurrence (if insufficient, fill in
03 30	00		02				*		FFFFFFFFH) (Last 2) Programming record (same as above)
			...						...
			0A						(Last 10 times) Programming record (same as
03 30	01		00	XXXXXX	3 times*				above) Total number of times
03 30	01		01	YYMMDDhhmmss	6		*		the meter is cleared (Last 1 time) Meter
				C0C1C2C3	4				clearing record:
				XXXXXX.XX	4	kWh			Occurrence time
				XXXXXX.XX	4	kWh			Operator code Before meter clearing Forward
				XXXXXX.XX	4	1.5 oz.			active energy Before meter clearing Reverse
				XXXXXX.XX	4	1.5 oz.			active energy Before meter clearing First quadrant reactive
				XXXXXX.XX	4	1.5 oz.			energy Before meter clearing Second quadrant reactive
				XXXXXX.XX	4	1.5 oz.			energy Before meter clearing Third quadrant reactive
				XXXXXX.XX	4	1.5 oz.			energy Before meter clearing Fourth quadrant reactive
				XXXXXX.XX	4	kWh			energy Before meter clearing A phase forward active
				XXXXXX.XX	4	kWh			energy Before meter clearing A phase reverse active
				XXXXXX.XX	4	1.5 oz.			energy Before meter clearing A phase first quadrant reactive
				XXXXXX.XX	4	1.5 oz.			energy Before meter clearing A phase second quadrant
				XXXXXX.XX	4	1.5 oz.			reactive energy Before meter clearing A Phase A third
				XXXXXX.XX	4	1.5 oz.			quadrant reactive energy meter before clearing Phase A
				XXXXXX.XX	4	kWh			fourth quadrant reactive energy meter before clearing
				XXXXXX.XX	4	kWh			Phase B forward active energy meter before clearing
				XXXXXX.XX	4	kWh			Phase B reverse active energy meter before clearing Phase
				XXXXXX.XX	4	1.5 oz.			B first quadrant reactive energy meter before clearing Phase
				XXXXXX.XX	4	1.5 oz.			B second quadrant reactive energy meter before clearing
				XXXXXX.XX	4	1.5 oz.			Phase B third quadrant reactive energy meter before clearing
				XXXXXX.XX	4	kWh			Phase B fourth quadrant reactive energy meter
				XXXXXX.XX	4	kWh			before clearing Phase C forward active energy meter
				XXXXXX.XX	4	1.5 oz.			before clearing Phase C reverse active energy meter before clearing Phase C first quadrant reactive energy

Data Identification				Data Format	data length (byte)	unit	Function		Data item name
D13 D12	D11 D10						Read	Write	
				XXXXXX.XX XXXXXX.XX XXXXXX.XX	4 4 4	1.5 oz. 1.5 oz. 1.5 oz.			Reactive energy in the second quadrant of phase C before the meter is reset Reactive energy in the third quadrant of phase C before the meter is reset Reactive energy in the fourth quadrant of phase C before the meter is reset
03	30	01	02 ... 0A				*		(Last 2 times) Electricity meter reset record ... (Last 10 times) Electricity meter reset record
03	30	02	00	XXXXXX	3 times*				Total number of demand resets
03	30	02	01	YYMMDDhhmmss C0C1C2C3 XX.XXXX YYMMDDhhmm	6 4 3 5		*		(Last) Demand reset record: 202 bytes Time of occurrence Operator code The total maximum forward active demand and the time of occurrence before the demand is cleared Total maximum reverse active demand and occurrence time before demand is cleared The maximum total reactive power demand in the first quadrant and the time of occurrence before the demand is cleared The total maximum reactive power demand in the second quadrant and the time of occurrence before the demand is cleared The total maximum reactive power demand in the third quadrant and the time of occurrence before the demand is cleared The total maximum reactive power demand in the fourth quadrant and the time of occurrence before the demand is cleared Maximum forward active demand of phase A and time of occurrence before demand is cleared The maximum reverse active demand in phase A and the time of occurrence before the demand is cleared Maximum reactive power demand in the first quadrant of phase A and its occurrence time before demand is cleared The maximum reactive power demand in the second quadrant of phase A and the time of occurrence before the demand is cleared Maximum reactive power demand in the third quadrant of phase A and its occurrence time before demand is cleared Maximum reactive power demand in the fourth quadrant of phase A and its occurrence time before demand is cleared Maximum forward active demand of phase B and time of occurrence before demand is cleared The maximum reverse active demand of phase B and the time of occurrence before the demand is cleared Maximum reactive power demand in the first quadrant of phase B and its occurrence time before demand is cleared Maximum reactive power demand in the second quadrant of phase B and its occurrence time before demand is cleared Maximum reactive power demand in the third quadrant of phase B and its occurrence time before demand is cleared Maximum reactive power demand in the fourth quadrant of phase B and its occurrence time before demand is cleared Maximum forward active demand of phase C and time of occurrence before demand is cleared Maximum reverse active demand in phase C and the time of occurrence before demand is cleared Maximum reactive power demand in the first quadrant of phase C and its occurrence time before demand is cleared Maximum reactive power demand in the second quadrant of phase C and its occurrence time before demand is cleared Maximum reactive power demand in the third quadrant of phase C and its occurrence time before demand is cleared Maximum reactive power demand in the fourth quadrant of phase C and its occurrence time before demand is cleared
03	30	02	02 ... 0A				*		(Last 2 times) Demand reset record ... (Last 10 times) Demand reset record
03	30	03	00	XXXXXX	3 times*				Total number of event clears (including total clears and item clears)
03	30	03	01	YYMMDDhhmmss C0C1C2C3 XXXXXXXX	6 4 4		*		(Last) Event clear record: Time of occurrence Operator code Event clear data identification code
03	30	03	02 ... 0A				*		(Last 2) Event clear record (same as above) ... (Last 10) Event clear record (same as above)
03	30	04	00	XXXXXX	3 times*				Total number of calibration times
03	30	04	01	C0C1C2C3 YYMMDDhhmmss YYMMDDhhmmss	4 6 6		*		(Last time) time calibration record: Operator code Time before calibration Time after calibration
03	30	04	02 ... 0A				*		(Last 2 times) Time calibration record (same as above) ... (Last 10 times) Time calibration record (same as above)

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12	D11 D10						Read	Write	
03 30	0D		00	XXXXXX	3 times*				Total number of times the
03 30	0D		01	YYMMDDhhmmss YYMMDDhhmmss XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX XXXXXX.XX	6 6 4 4 4 4 4 4 4 4 4 4 4 4 4	 kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz. kWh kWh 1.5 oz. 1.5 oz. 1.5 oz. 1.5 oz.	*		meter cover was opened (last 1 time) Meter cover opening record: Occurrence time End time Total forward active energy before meter cover opened Total reverse active energy before meter cover opened Total reactive energy in the first quadrant before meter cover opened Total reactive energy in the second quadrant before meter cover opened Total reactive energy in the third quadrant before meter cover opened Total reactive energy in the fourth quadrant before meter cover opened Total forward active energy after meter cover opened Total reverse active energy after meter cover opened Total reactive energy in the first quadrant after meter cover opened Total reactive energy in the second quadrant after meter cover opened Total reactive energy in the third quadrant after meter cover opened Total reactive energy in
03 30	0D		02 ... 0A				*		the fourth quadrant after meter cover opened (last 2 times) Meter cover opening record (same as above) ... (Last 10 times) Opening record (same as above) Total
03 30	0E		00	XXXXXX	3 times*				number of opening button box
03 30	0E		01 ... 0A				*		(Last 1) Open button box record (same as opening the cover) ... (Last 10 times) Open button box record (same as opening the cover)

Table A.5 Parameter data identification coding table

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12	D11 D10						Read	Write	
04	00	01	01	YYMMDDWW	4	Year Month Day Week	*	*	Date and day of the week (where 0 represents Sunday),
			02	hhmmss	3	Hour Minute	*	*	time,
			03	NN	1		*	*	maximum demand, cycle
			04	NN	1		*	*	slip time, meter
			05	XXXX	2	Second	*	*	calibration pulse width,
			06	YYMMDDhhmm	5	Minute Millisecond	*	*	switching time for two time zone tables,
			07	YYMMDDhhmm	5	Year Month Day Hour	*	*	switching time for two daily time period
04	00	02	01	NN	1		*	*	tables, number of time zones
			02	NN	1		*	*	per year (py14), number of daily
			03	NN	1		*	*	time period tables (qy8), number of daily time periods
			04	NN	1		*	*	(number of daily
			05	NNNN	2		*	*	switching), my14, number of tariffs
04	00	03	01	NN	1		*	*	(ky63), number of public holidays
			02	NN	1		*	*	(ny254), number of
			03	NN	1		*	*	automatic cycle displays, time
			04	NN	1		*	*	displayed per screen, number of decimal places for
			05	NN	1	Year Month Day Hour Minute Second digit display	*	*	displayed energy, number of decimal
04	00	04	01	NNNNNNNNNNNN	6		*	*	places for
			02	NNNNNNNNNNNN	6		*	*	displayed
			03	NN...NN	32		*	*	power (maximum demand), number of key cycle
			04	XXXXXXXXXXXX	6		*	*	displays, communication address,
			05	XXXXXXXXXXXX	6		*	*	meter number, asset management code (ASCII code), rated voltage (ASCII code) rated current/basic current

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13	D12	D11	D10				Read	Write	
			06	XXXXXXXXXXXX	6	imp/kWh imp/kvarh	*		Maximum current (ASCII code)
			07	XXXXXXXX	4		*		Active power accuracy level (ASCII code)
			08	XXXXXXXX	4		*		Reactive power accuracy level (ASCII code)
			09	XXXXXX	3		*		Meter active power
			0A	XXXXXX	3		*		constant Meter reactive
			0B	XX...XX	10		*		power constant Meter model (ASCII
			0C	XX...XX	10		*		code) Production date (ASCII code)
			0D	XX...XX	16		*		Protocol version number (ASCII code)
04	00	05	01	XXXX	2		*		Meter operation status word 1
			...	...	...		...		...
			07	XXXX	2		*		Meter operating status word 7 Meter
			FF				*		operating status word data block Active power
04	00	06	01	NN	1		*	*	combination mode Characteristic word
			02	NN	1		*	*	Reactive power combination mode 1 Characteristic
			03	NN	1		*	*	word Reactive power combination mode 2
04	00	07	01	NN	1		*	*	Characteristic word Modulated infrared optical port communication rate
			02	NN	1		*	*	Characteristic word Contact infrared optical port communication rate
			03	NN	1		*	*	Characteristic word Communication port 1 communication
			04	NN	1		*	*	rate Characteristic word Communication port 2
			05	NN	1		*	*	communication rate Characteristic word Communication
04	00	08	01	NN	1		*	*	port 3 communication rate
			02	NN	1		*	*	Characteristic word Weekend characteristic word
04	00	0B	01	Human Rights	2	day	*	*	Weekend day time period meter
			02	Human Rights	2	time	*	*	number used 1st settlement day of
			03	Human Rights	2	day time	*	*	each month 2nd settlement day of
04	00	0C	01	NNNNNNNN	4			*	each month 3rd settlement day of each month Level 0 password
			...	...	...			...	...
			0A	NNNNNNNN	4			*	Level 9
04	00	0E	01	NN.NNNN	3	kW	*	*	password: Forward active power
			02	NN.NNNN	3	kW	*	*	upper limit, Reverse active power
			03	NN.N	2	ln	*	*	upper limit, Voltage
			04	NN.N	2	ln	*	*	upper limit, Voltage
04	01	00 00		MMDDNN	3		*	*	lower limit, First set of time zone
				...	...		...	...	table data: First time zone start date and day time table number
				MMDDNN	3		*	*	The starting date and daily time table number of the 14th time
04	01	00	01	hhmmNN	3		*	*	zone The first set of daily time table data: The
				...	...		...	...	starting time and rate number of the first time period
				hhmmNN	3		*	*	The start time and rate number of the 14th period,
04	01	00 02					*	*	the first set of the 2nd day period table data
			...				...	...	...
			08				*	*	The first set of data for the 8th day time period table. The
04	02	00 00		MMDDNN	3		*	*	second set of data for the time zone table:
				...	...		...	...	The starting date and day time period table number of the 1st time zone.
				MMDDNN	3		*	*	The starting date and daily time table number of the 14th time
04	02	00	01	hhmmNN	3		*	*	zone The second set of daily time table data:
				...	...		...	...	The starting time and rate number of the first time period
				hhmmNN	3		*	*	The start time and rate number of the 14th period, the
04	02	00 02					*	*	second set of the second day period table data
			...				...	...	...
			08				*	*	The second set of 8th day time table data

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Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12 D11 D10							Read	Write	
04	03	00	01	YYMMDDNN	4		*	*	1st Public Holiday Date and Time Table
			...	...			...	...	...
			FE	YYMMDDNN			*	*	No. 254 Public holiday date and time table number automatically
04	04	01	01	NNNNNNNN	4		*	*	cycle display 1st screen display data item
			...	...			...	...	...
			FE	NNNNNNNN			*	*	Automatically cycle through the 254th screen display data item.
04	04	02	01	NNNNNNNN	4		*	*	Press the button to cycle through the 1st screen display data item.
			...	...			...	...	...
			FE	NNNNNNNN			*	*	Press the key to cycle through the display data items on the 254th
04	80	00	01	NN...NN	32		*		screen: Manufacturer software version number (ASCII
			02	NN...NN	32		*		code) Manufacturer hardware version number (ASCII
			03	NN...NN	32		*		code) Manufacturer serial number (ASCII code)
Note 1: The starting value of the daily time table number and rate number is 1. Note 2: If the time zone table data is insufficient, set the time zone number to fill in the last time zone data. If the daily time table data is insufficient, set the daily time period number to fill in the last daily time period data. If the holiday data is insufficient, set the number of public holidays to fill in the last public holiday data. Note 3: For data items transmitted in ASCII, NUL is added after the missing bytes. Note 4: It is recommended to use the company code as the manufacturer number. Note 5: If the monthly settlement date value is 9999, it means that this settlement date has not been set. Note 6: In the loop display setting, NNNNNNNN represents the data identifier corresponding to each display item.									

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Appendix B (Electricity Measurement Standard) No. 645-09-001/002 "Multi-function Electricity Meter Communication Protocol" Record Document

Table 1 Electric energy data identification code table

Data Identification				Data Format	data length (byte)	unit	Function		Data item name
D13	D12	D11	D10				Read	Write	
00	90	01	00	XXXXXX.XX	4	kWh	*		(Current) remaining battery power
			01	XXXXXX.XX	4	kWh	*		(Current) overdrawn power
00	90	02	00	XXXXXX.XX	4	Yuan	*		(Current) remaining amount
			01	XXXXXX.XX	4	Yuan	*		(Current) overdraft amount
00	0B	00	00	XXXXXX.XX	4	kWh	*		Total accumulated active power consumption of the current settlement period
00	0B	00	01	XXXXXX.XX	4	kWh	*		Total cumulative active power consumption of the previous settlement period

Table 2 Variable data identification coding table

Data Identification				Data Format	data length (byte)	unit	Function		Data item name
D13	D12	D11	D10				Read	Write	
02	80	00	0B	XXXX.XXXX	4	kWh	*		Current tiered electricity prices

Table 3 Event record data identification code table

1D 00 00 01				XXXXXX	3		*		* Number of trips
1D 00 01		01		YYMMDDhhmmss	6		*		(Last) trip time
		02		C0C1C2C3	4		*		(Last time) Trip operator code
		03		XXXXXX.XX	4	kWh	*		Total forward active energy at the time of tripping (last time)
		04		XXXXXX.XX	4	kWh	*		Total reverse active energy when tripping (last time)
		05		XXXXXX.XX	4	1.5 oz.	*		Total reactive energy in the first quadrant at the time of (last trip) trip
		06		XXXXXX.XX	4	1.5 oz.	*		Total reactive energy in the second quadrant at the time of (last trip) trip
		07		XXXXXX.XX	4	1.5 oz.	*		Total reactive energy in the third quadrant at the time of (last trip) trip
		08		XXXXXX.XX	4	1.5 oz.	*		Total reactive energy in the fourth quadrant at the time of (last trip) trip
		FF					*		(Last) trip record data block
1D 00 2Z 02			...				*		(Last 2) Trip record (same as above)
		0A							...
		FF							(Last 10 times) Trip record (same as above)
									A data block of 10 records for a certain item
1E 00 00 01				XXXXXX	3		*		* Closing times
1E 00 01		01		YYMMDDhhmmss	6		*		(Last) Closing time
		02		C0C1C2C3	4		*		(Last time) Closing operator code
		03		XXXXXX.XX	4	kWh	*		Total forward active energy when closing (last time)
		04		XXXXXX.XX	4	kWh	*		(Last time) Total reverse active energy when closing

		05		XXXXXX.XX	4	1.5 oz.	*		(Last time) Total reactive energy in the first quadrant when closing
		06		XXXXXX.XX	4	1.5 oz.	*		(Last time) Total reactive energy in the second quadrant when
		07		XXXXXX.XX	4	1.5 oz.	*		closing (Last time) Total reactive energy in the third quadrant
		08		XXXXXX.XX	4	1.5 oz.	*		when closing (Last time) Total reactive energy in the fourth
		FF					*		quadrant when closing (Last time) Closing
1E 00	ZZ 02		...				*		record data block (Last 2 times) Closing record (same as above)
			0A						...
			FF						(Last 10 times) Closing record (same as above)
									10 record data blocks of a certain item
Note 1: The electricity purchase date is the time when the pre-purchased electricity quantity (amount) is entered into the meter. Note 2: When the local automatic tripping and closing is performed, the operator code in the tripping and closing records is FFFFFFFF.									

Table 4 Parameter data identification coding table

Data Identification				Data Format	Data length (bytes)	unit	Function		Data item name
D13 D12	D11 D10						Read	Write	
04 00		03	06	NNNNNN	3		*	*	Current transformer ratio
			07	NNNNNN	3		*	*	Voltage transformer ratio